Information visualization

Link to our git project : https://github.com/AmirBourry/Songs-All

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Users

The project targets music industry analysts, researchers, and enthusiasts who want to explore patterns and trends in the music industry, particularly focusing on:

- Release timing of albums and songs
- Geographic distribution of artists
- Genre evolution over time
- Explicit content trends
- Artist productivity and career timelines

Visual Tasks and Goals

1. **Geographic Distribution Analysis**
    - Visualize countries with high concentrations of famous artists
    - Compare artist popularity across different regions
    - Identify geographical trends in the music industry
2. **Temporal Pattern Analysis**
    - Track album release patterns over time
    - Visualize artist productivity through career timelines
    - Analyze genre evolution across years
    - Identify seasonal patterns in album releases
3. **Genre Analysis**
    - Compare genre proportions in different years
    - Track genre popularity trends over time
    - Analyze genre distribution by country
4. **Content Analysis**
    - Track explicit content trends over time
    - Compare explicit content usage across different genres
    - Analyze correlation between explicit content and classic status

Description

We begin on a page where you select a country and you have a scatter plot that appears with the proportion of deezer fans, for each genre of that country. On the next page we keep that country and you enter a year, a calendar appears with a visualization of when albums were released for that year and country. You can change the country and year if you have no data. You can click on a date and the list of albums and the artists will appear, you can click on the artist and a timeline of said person will appear. You can see when songs and albums were released for that artist and in what year. You can then click on the link to go to the next page that takes into account that specific year (of the calendar) and you will see a pie chart representing the proportion of genre for that year. You can modify the year if you like. The final page is with the year you visualized on the pie chart, it will represent a bubble chat showing if the albums of that year have won awards versus the explicitness. You can also visualize the genre of the album and it’s popularity.

Raw Data

For our project we want to use tree structures for our data.

Table	Raw Attributs
Album	- *explicitLyrics:* Has explicit content - *title:* Album's title - *length: Album's length* - *id_artist*: Unique identifier of the artist who released the album. - *_id*: Unique identifier of the album. - *name*: Name of the artist associated with the album. - *genre*: Musical genre of the album (e.g., rock, pop, jazz). - *dateRelease*: Release date of the album (including year, month, and day). - *cover*: Image of the album's cover. - *country*: County of origin of the album. - *deezerFans*: Number of Deezer fans for the album, representing their popularity.
Artist	- *_id*: Unique identifier of the artist. - *name*: Artist's name. - *recordLabel*: Music label under which the artist released music. - *country*: Country of origin of the artist. - *deezerFans*: Number of Deezer fans for the artist, representing their popularity. - *genre*: The primary musical genre of the artist (e.g., pop, rock, jazz). - *locationInfo* :
Song	- *_id:* Unique identifier of the song. - id_album: Unique identifier of the album that the song is in. - *genre:* Song genre.

	- *releaseDate:* Song's release date. - *explicitLyrics:* Has explicit content. - *isClassic:* Does it have an award such as the hall of fame, diamond, platinum, gold or hot 100

Visualization Techniques and Implementers

Valentin : Pie Chart Visualization

The technique used will be a **pie chart** to show the proportion of album releases across different genres for a selected year. The pie chart is effective at visualizing parts of a whole, which is the visual goal in this case.

List of (Raw) Attributes Needed from the WASABI Dataset

In song :

- genre
- releaseDate

Informal Description of Processing Raw Data

Filter by Year: The page already loads with the year from the previous page, and we allow users to select a specific year. Based on the year selection, we will filter the data to include only the releases from that year.
Group by Genre: Once filtered, the dataset will be grouped by the `genre` attribute and mapping it to a broader genre using subgenres_by_genre.json, to count the number of releases in each genre for the selected year.
Count Releases: For each genre, the number of album releases (distinct by `_id`) will be calculated to ensure accurate representation of genre popularity.
Normalize Percentages: After counting releases for each genre, the data will be normalized into percentages so that they can be visualized as proportional segments of a pie chart.

Visual Mapping of Variables

- **Genre** (Categorical Variable)
  → **Color**: Each genre will be represented by a different color in the pie chart. We are regrouping subgenres into genres (exemple : Alternative Rock, Punk Rock into Rock).
- **Count of Releases** (Numerical Variable)
  → **Angle** (Slice Size): The number of releases in each genre will determine the angle of each slice in the pie chart.
  → **Text** : The number of percentages the slice of genre is taking is displaying when hovering a part of the pie chart.
- **Year** (Temporal Variable)
  → **Text** : The year of release will be mapped to an input to allow users to change the year and update the pie chart accordingly.

https://datavizcatalogue.com/methods/pie_chart.html

Carolina : Bubble Chart Visualization

The visual goal is to display information, in this case whether an album won an award (“classiness”) vs. the use of explicit language (“explicitness”) in this album and compare these scores over time and between different music genres. A bubble chart is used to achieve this. Each bubble represents an album and contains information on the explicitness, the classiness, the number of fans and genre of the respective album. By hovering over one of the bubbles, a tooltip shows the title of the album and the name of the artist. It is possible to change the displayed year with a filter to get an idea of the development of classiness and explicitness over time. Comparing the scores of albums of different genres is possible by hovering over the desired genre in the legend. This will highlight all albums of this genre.

List of (raw) attributes from the WASABI dataset

- Album
  - `dateRelease`
  - `explicitLyrics`
  - `genre`
  - `title`
  - `name`
  - `deezerFans`
- Song
  - `id_album`
  - `explicitLyrics`
  - `isClassic`

Informal description of the processing of raw data

1. For every album in a specific year, `explicitness` of an album will be calculated as the proportion of songs in the album that are marked as `explicitLyrics`.
2. For every album in a specific year, `classicSongs` will be calculated as the proportion of songs in the album that are marked as `isClassic`.
3. With the information `explicitness`, `classicSongs`, `deezerFans` and `genre`, a bubble can be created for every album.
4. The tooltip shows `title` and `name` (artist name) of the album.

Visual mapping of variables

- `classiness` (numerical variable) as position (x-axis)
- `explicitness` (numerical variable) as position (y-axis)
- `deezerFans` (numerical variable) as size (area of the bubble)
- `genre` (categorical variable) as color of the bubble

Clervie: Calendar Visualization

Description

This code implements an interactive calendar visualization that displays album release dates for a specific country and year. The visualization features:

- A country selector dropdown
- A year input field
- A 12-month calendar view arranged in 3 rows of 4 months
- Color-coded days indicating album releases
- Interactive tooltips showing album and artist information
- Click functionality to pin album information for specific dates
- Visualization of the artists timeline (another visualization)
- Navigation to a second page for additional analysis

List of (raw) attributes from the WASABI dataset

- Album:
  - `title`: Album title
  - `name`: Artist name
  - `dateRelease`: Release date
  - `date`: Formatted date
  - `Country code` (derived from file structure)

Informal description of the processing of raw data

1. Data Structure Processing
   - Albums are organized in a hierarchical structure: year → country → files
   - Data is loaded from JSON files stored in */data/processed_albums/${year}/${countryCode}/${file}*
   - Albums are grouped by release dates for efficient lookup

2. Date Processing
   - Dates are standardized into YYYY-MM-DD format
   - Partial dates (YYYY or YYYY-MM) are padded with default end-of-period values
   - Leap year calculations are performed for February

3. Data Aggregation
   - Multiple albums released on the same date are grouped together
   - Album information is stored in a map structure for quick access
   - Country codes are extracted and sorted alphabetically

Visual mapping of variables

1. Temporal Variables
   - Calendar grid: Represents days of each month
   - Month headers: Display month names
   - Day cells: Show day numbers

2. Categorical Variables

- Color coding:
    - Light gray: No releases
    - Green (#5fd964): Has releases
- Country selector: Dropdown of available countries

3. Interactive Elements
- Hover effects: Shows album information in tooltip
- Click interaction: Pins album information
- Click interaction on the artist : Gets visualization of artists timeline
- Day cells: Interactive elements for showing release information
- Links: Artist names are clickable and lead to artist pages

Conclusion

This visualization effectively presents temporal patterns in album releases through an intuitive calendar interface. Key strengths include:

1. Usability
- Clear visual hierarchy
- Intuitive interaction patterns
- Easy filtering by country and year

2. Information Design
- Efficient use of color coding
- Clear distinction between days with and without releases
- Detailed information available on demand
- Organized presentation of multiple releases per day

3. Technical Implementation
- Efficient data structure for quick lookups
- Proper handling of edge cases (leap years, partial dates)
- Smooth interactions and transitions
- Modular code structure

The visualization successfully achieves its goal of showing temporal patterns in album releases while providing detailed information through interactive elements. The implementation balances information density with usability, making it an effective tool for exploring music industry release patterns.

Amir : Choropleth Map Visualization

Visualization of Countries on a Choropleth Map

The visualization technique used is a choropleth map, which allows for the display of countries based on their number of famous artists. The more famous artists a country has, the darker the shade of red on the map. The fewer artists, the closer the color is to white. This visualization helps users quickly identify regions of the world with a high density of popular artists. A legend accompanying the map indicates the colors corresponding to the number of famous artists.

List of (raw) attributes from the Deezer dataset

Artist:

- _id: Unique identifier of the artist.
- name: Name of the artist.
- country: Country of origin of the artist.
- deezerFans: Number of Deezer fans for the artist, representing their popularity.
- genre: The primary musical genre of the artist (e.g., pop, rock, jazz).

Informal description of the processing of raw data

Filtering and selection: Artist data is filtered by country and popularity (based on the number of Deezer fans) to retain only artists with a certain level of fame. Key information such as the artist's name (*name*), country (*country*), number of Deezer fans (*deezerFans*), and musical genre (*genre*) are extracted.

Data aggregation: The total number of famous artists is calculated for each country based on the number of artists associated with it. This total is then used to color each country on the map.

Adding additional information: Each country on the map can be associated with a tooltip displaying extra information such as the most popular artists (*name*), their total number of fans (*deezerFans*), and the most common musical genre (*genre*) in that country.

Visual mapping of variables

- Number of famous artists per country: The number of artists is mapped to a color on the map, ranging from white (few artists) to red (many artists). The total number of artists per country is calculated by summing up all the famous artists associated with each country.
- Color (representing artist density): The density of famous artists per country is represented by a color palette, with red symbolizing countries with the highest number of famous artists, while countries with few artists trend toward white.
- Tooltip: When users hover over or click on a specific country, a tooltip will display details such as the names of the most popular artists, their Deezer fan count, the total number of famous artists, and the primary musical genre associated with the country.

This visualization allows users to quickly identify countries with a high concentration of famous artists, spot geographical trends in the music industry, and compare artist popularity and genres across different nations. The choropleth map is a powerful tool for understanding the global distribution of popular artists and their regional influence.

Chloé: Timeline Visualization

The visualization technique is a timeline to visualize over the years the release dates of the albums and songs of an artist. The visual goal is to display the frequency of an artist's releases over time. This will help in understanding trends, such as periods of high or low productivity.

List of (raw) attributes from the WASABI dataset:

Artist:

- **`_id` :** Unique artist id
- **`Name` :** Artist's name

Album:

- **`_id` :** Album's id
- **`Id_artist` :** artist's id
- **`dateRelease`:** Album's release date
- **`Title` :** Album's title
- **`Genre` :** Album's genre
- **`Cover` :** Album's cover

Song:

- **`_id` :** Song's id
- **`id_album` :** Song's album's id
- **`title` :** Song's title
- **`Genre` :** Song's genres
- **`releaseDate` :** Song's release date

Informal description of the processing of raw data

- **Link the artist with its albums and songs:** From the calendar, we can get an artist's name from which we can retrieve the artist's id. This allows us to fetch all the albums and songs that were released by that artist.
- **Date Formatting**: Ensure all release dates are in a consistent format (e.g., YYYY-MM-DD) to accurately map them chronologically on the timeline.
- **Track list of the album:** As I want to display the album's information while hovering the album's dot, I want to link the album to its track list. However, we do not have this attribute in the dataset but in the songs' dataset we have the id_album attribute with which we can link to the album.

Visual mapping of variables

- **Green Circles :** Identify the artist's album releases.
- **Size of green circles :** The size varies with the amount of songs in the album.
- **Pink/Purple triangles :** Identify the artist's song releases.
- **Release dates on a timeline :** Temporal variable. Maps the release dates of albums and songs chronologically. Each point on the timeline corresponds to a release date (X-axis)

- **White circles :** Appears when we can find several releases on the same day. As it wasn’t possible to see that there were several points on the same date, it was important to have a visible element to represent that there are several elements that were released on that date.
- **Pop-up appearing when hovering a point on the timeline :**
    - **For albums :** Display of the title, the date, the genre, the cover and the track list.
    - **For songs :** Display of the title, the date and the genre.
    - **For group circles :** Display of the date, the list of albums (with title, genre and cover) and the list of songs (with title).

This timeline visualization effectively highlights the artistic journey of an artist by mapping out the release dates of albums and songs over the years. By using distinct shapes and colors to differentiate between albums and individual song releases, users can easily spot periods of high or low productivity. The varying sizes of the circles, based on the number of songs per album, add another insight, making it simple to identify more substantial releases at a glance.

The interactive elements, such as hover pop-ups displaying detailed information (like album covers, genres, and track listings), enrich the user experience by providing context without obstructing the visual. Additionally, the use of white circles to signify multiple releases on the same date ensures clarity and prevents data overlap, enhancing readability.

Overall, this visualization provides a comprehensive overview of an artist's creative output, enabling users to quickly discern trends, peak activity periods, and the evolution of their music over time. It's an intuitive tool for exploring an artist’s discography and understanding their artistic progression.

Furkan: Scatter Plot visualization

The aim of the visualization technique is to compare the popularities of different genres among the years for a specific country. For this, we are using a scatter plot. Before this visualization, the user will choose a country from the previous visualization. The scatter plot will indicate the data from that country. The x-axis represents the years, the Y-axis represents the popularity (number of fans), and each data point represents different albums from that country. In order to distinguish the different genres, different colors are used to represent the albums. With this visualization, users will obtain knowledge about the genre trends of that country over the years. Also users can reach specific information about the album.

List of (raw) attributes from the WASABI dataset:

Album:

`title:` ***Album title***

`genre:` ***Album genre***

`country:` ***Place where the album was released***

`dateRelease:` ***Release date***

`deezerFans:` ***Number of Deezer fans***

Informal description of the processing of raw data

Filtering by country: Since the scatter plot is for one country, only albums that are from the selected country are shown. A dropdown menu is used to select a country. For the albums whose country field is empty, “Unknown” field is created. All of these albums are reachable in the dropdown menu with the “Unknown” option.
Genre classification: Since there are lots of different genres, for the classification, subgenres_by_genre.json is used in order to map the genres to broader subgenres. That broader subgenres are used directly in the visualization. Each genre has a different color in the visualization.
Additional Informations: The Scatter Plot gives a broad information about the country. Additional information about the album is viewed when a user hovers a specific data point in the scatter plot.

Visual mapping of variables
Color: The color is mapped to the Genre field from the dataset. This allows us to distinguish and identify different genres easily. Among all visualizations(even between other visualizations), representation is consistent. The same color is used to represent the same

genre. With this way each color is uniquely representing one genre and it is memorizable by the user.
Popularity: In the Y axis, the values represent the number of Deezer Fans field for each album. The concentration of the same color on the same side of that scale gives broad information about trends. In the plot, linear scale is used; therefore, it is easy to distinguish the more popular albums and genres from others.
Year: In the X axis the year represents the release date for each album. Since the dataset does not include the month and exact day, the mapping is one on one. From left to right, in chronological order.
Country: To select a country, A dropdown menu is used. This way, only albums belonging to that country are shown. Countries are shown with their country code in the dropdown menu.
Title: The name of the album, belonging country and the exact number of Deezer Fans are mapped to an information tag, which is triggered while hovering a specific data point in the scatter plot. This way, users can see exact basic details about the album.

The main reason for choosing this visualization technique was it was so compatible with other techniques and meets perfectly with our requirements. It was so compatible with our previous visualization. We had a Country Map before this visualization, and the plan was selecting a country from the Country Map, which will result with the scatter plot of that specific country. Unfortunately, because of the absence of that visualizer, we did not succeed with that combination. Therefore just a dropdown selection is used to work with a similar purpose.